IILM UNIVERSITY
GREATER NOIDA, UTTAR PRADESH

School of Computer Science and Engineering
IILM University, Greater Noida, Uttar Pradesh

ACADEMIC INTERNSHIP PRESENTATION | WEB DEVELOPMENT

CLAIM PRO

Insurance Claims Management System

A responsive, high-performance Single-Page Application (SPA) architected to streamline the full lifecycle of insurance claims—including multi-step intake filing, document verification, officer audit workflows, approval gating, financial settlement disbursement, and actuarial reporting.

CORE IMPLEMENTATION STACK (PURE CLIENT-SIDE ARCHITECTURE):
HTML5 Semantic Standards • CSS3 Design System • Vanilla JavaScript (ES6+) • Browser LocalStorage Engine

CANDIDATE CREDENTIALS

Shreshth Chauhan

University Roll No.: **2410030045**
Class Section: **3CSE14**
Course / Degree: **B.Tech Computer Science & Engg.**
Academic Batch: **2024–2028 (Session 2026–2027)**
Host Organization: **Data Alcott Systems / FreeInternships.in**
Internship Student Code: **DAS012321**
Internship ID / Task: **5696 | Portal Task #5110**
Internship Duration: **21 Oct 2026 – 28 Oct 2026**

Project Links: GitHub: github.com/shresthchauhan14-cpu/claim-pro | Live Demo: shresthchauhan14-cpu.github.io/claim-pro | YouTube: youtu.be/Gdbz637qS1E

Academic Credentials & Internship Overview

Verified student enrolment records and institutional internship placement metadata.

CANDIDATE ACADEMIC PROFILE

Student Full Name

Shreshth Chauhan

University Roll Number

2410030045

Class Section

3CSE14

Degree & Program

B.Tech in Computer Science and Engineering

Academic Batch

2024–2028 (Current Session: 2026–2027)

Academic Department

School of Computer Science & Engineering

Institution / University

IILM University, Greater Noida

Campus Location

Plot No. 6, 7 & 8, Knowledge Park II, Greater Noida, UP

INTERNSHIP PLACEMENT PROFILE

Host Organization

Data Alcott Systems / FreeInternships.in

Internship Track / Domain

Web Development (Virtual Internship Track)

Student Registration Code

DAS012321

Internship Portal Identifier

Internship ID: 5696 | Portal Task ID: #5110

Internship Duration

21 October 2026 – 28 October 2026 (1 Week Intensive)

Official Certificate ID

DAS-COMP-252/2026 (Verified Online)

Supervising HR & Manager

Gokul Rajan (Reporting Mgr) & Thirumalai Kumar (HR)

Core Engineering Focus

Pure Vanilla SPA Architecture & Actuarial State Management

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Data Alcott Systems & FreeInternships.in

Institutional profile, technical domain standards, and internship training objectives.

Corporate Profile & Mission

Industry-Oriented Virtual Internship: Data Alcott Systems is an IT technical consulting and skill development organization operating the FreeInternships.in portal to bridge academic theory and real-world software engineering.

Practical Project-Based Evaluation: Interns are assigned comprehensive, domain-specific tasks evaluated against industry standards of code architecture, code hygiene, and user experience.

Virtual Learning Environment: Provides remote engineering tracks for students across India, fostering autonomous problem-solving, documentation discipline, and production readiness.

Web Development Track Standards

Zero Framework Crutch: Requires students to implement real-world enterprise interfaces from scratch using vanilla HTML5, CSS3, and ES6+ JavaScript without reliance on heavy frameworks.

Rigorous Business Logic: Mandates complete lifecycle execution—from multi-step user onboarding and input validation to underwriting audits and settlement disbursement.

State & Storage Engineering: Emphasizes robust client-side storage architectures utilizing browser LocalStorage with error boundaries and data backup/restore capabilities.

Evaluation & Certification Criteria

Portal Task Evaluation (#5110): Project deliverables—including responsive layouts, clean code structure, and automated unit testing—are reviewed by technical managers.

Verified Credential Award: Successful candidates receive a cryptographically verifiable completion certificate with an online validation URL and individual serial number.

Industry Mentorship: Technical mentorship and HR oversight provided by Gokul Rajan (Reporting Manager) and Thirumalai Kumar (HR Operations).

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Introduction & Project Purpose

Modernizing insurance claim operations with a responsive, client-side digital workspace.

The Mission of CLAIM PRO

Eliminating Paper Inefficiencies

- Re-engineers the traditional insurance claims lifecycle into an intuitive, responsive Single-Page Application (SPA).
- Replaces cumbersome paper forms, physical bill submission, and disconnected spreadsheets with a unified digital surface.
- Provides total status transparency to policyholders while granting claims adjusters powerful auditing and verification tools.
- Eliminates communication blackouts by delivering real-time lifecycle tracking from submission to bank settlement.

End-to-End Operational Lifecycle

5 Cohesive Functional Phases

- 1. Guided Intake Filing: 4-step wizard with real-time validation and multi-file drag-and-drop document upload.
- 2. Document Verification Audit: Granular inspection of hospital bills, FIRs, and identity proofs with audit status toggles.
- 3. Officer Approval Gating: Intercepts approvals to prevent payouts if any document remains unverified or rejected.
- 4. Financial Settlement: Payout limit enforcement, transaction ID generation, and printable payment vouchers.
- 5. Actuarial Telemetry: Real-time SVG KPI trends, status distributions, and formula-injection-safe CSV exports.

Engineering Philosophy

Zero Framework & Backend Overhead

- Built strictly with pure modern web standards: semantic HTML5, custom tokenized CSS3, and vanilla ES6+ JavaScript.
- Operates entirely on client-side state persistence using browser LocalStorage with structured JSON schema serialization.
- Guarantees instant UI responsiveness and sub-second execution without external backend server latency.
- Zero installation barrier: Evaluators and users can execute and test the complete application instantly in any modern web browser.

Problem Statement & Industry Inefficiencies

Analyzing the systemic operational failures in legacy, paper-driven insurance claim management.

1. Severe Turnaround Latency & Tracking Gaps

Weeks Lost in Manual Paper Routing

- Physical claim forms and bills require inter-office postal dispatch, manual data entry, and multi-desk routing.
- Claimants face zero visibility into processing stages, overwhelming support staff with repetitive status inquiries.
- Missing or illegible documents stall claims indefinitely without automated claimant notifications.

2. Fraud Risks & Unverified Approvals

Lack of Compliance Gating & Verification Audits

- Physical bills and receipts are vulnerable to forgery, duplicate claims, and altered financial values.
- Disjointed spreadsheets allow officers to authorize disbursements before supporting invoices are verified.
- Audit trails are incomplete or non-existent, creating immense compliance liability during insurance regulatory audits.

3. Data Silos & Absent Actuarial Telemetry

Disconnected Spreadsheets & Manual Reporting

- Actuaries and senior management cannot monitor exposure, reserves, or approval velocity in real time.
- Manual spreadsheet consolidation causes human computation errors and delayed loss ratio projections.
- Raw CSV exports frequently trigger spreadsheet formula injection vulnerabilities (=SUM, @cmd).

Core Engineering Objectives & Architecture Boundaries

Clear demarcation of implemented functional capabilities versus architectural boundary limits.

CORE ENGINEERING OBJECTIVES

1. Guided Intake Filing System: Deliver an intuitive 4-step filing wizard featuring real-time input validation (Indian phone numbers, email regex) and multi-file drag-and-drop document staging.

2. Granular Document Verification: Implement an audit interface allowing claims adjusters to inspect hospital bills, FIRs, and repair estimates, toggling verified or rejected status with full reason tracking.

3. Strict Officer Approval Gating: Build business logic guards that strictly prevent claim approval if any document is unverified or rejected, protecting insurers against unauthorized disbursements.

4. Financial Settlement Engine: Provide complete settlement reconciliation with transaction ID generation, dynamic deductible calculation, and printable disbursement vouchers.

5. Live Actuarial Telemetry: Render 10 dynamic KPI metric cards and SVG visual charts (trend lines, status donuts) calculating exposure, reserves, and velocity without external charting libraries.

ARCHITECTURAL BOUNDARIES & DEMARCATION

Client-Side Architecture Demarcation: CLAIM PRO is intentionally engineered as a pure frontend demonstration application. All data persistence, validation rules, and business logic execute entirely in browser memory and LocalStorage.

No External Server Database: Does not connect to external SQL databases (PostgreSQL, MySQL) or remote authentication backends (OAuth2, Firebase). This guarantees 100% portability and instant execution.

Simulated Banking Payouts: Settlement processing generates valid transaction identifiers and calculates exact net payouts, but does not initiate live electronic fund transfers via NEFT/RTGS or banking APIs.

Storage Quota Management: To operate cleanly within the 5MB browser LocalStorage limit, uploaded document files are converted to preview URLs and persisted as structured metadata entries rather than large base64 binaries.

System Architecture & Layered Data Flow

Zero-dependency engineering stack powered by custom reactive state management.

Layer 1: Semantic Presentation
HTML5 & Tokenized CSS3

HTML5 Semantic Structure: Built using strict semantic elements (<header>, <nav>, <main>, <section>, <dialog>) ensuring accessibility, clean DOM tree hierarchy, and fast layout rendering.

Custom Tokenized CSS3: Engineered with 40+ CSS custom properties (--color-primary, --border-radius, --space-md) supporting seamless light and dark mode theming without third-party frameworks like Tailwind or Bootstrap.

Responsive Grid & Flexbox: Fluid CSS grid dashboard layouts that scale effortlessly from 360px mobile displays to 4K ultra-wide monitors with zero layout shifts.

Layer 2: Reactive State Controller
Vanilla JavaScript (ES6+)

Custom Pub/Sub State Model: Centralized Store singleton implementing an EventEmitter pattern. Any mutation in claims triggers automated recalculations across all dashboard widgets.

Single-Page Router: Hash-based client-side routing engine enabling instantaneous view transitions between Dashboard, Claims, Audit, Settlement, Reports, and Settings without page reloads.

Data Hygiene & Security: Zero eval() or innerHTML execution for user input. Client-side sanitization guards protect against Cross-Site Scripting (XSS) and CSV formula injection.

Layer 3: LocalStorage Engine
Structured Browser Persistence

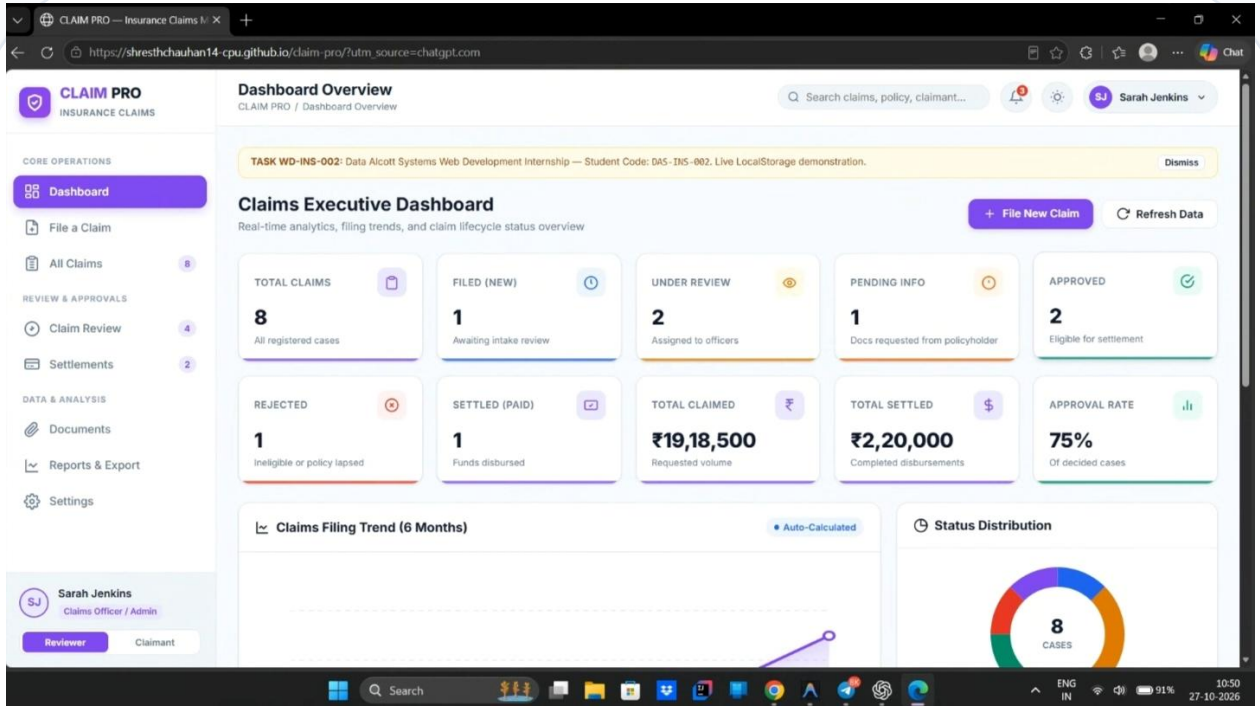
JSON Schema Serialization: Encapsulates claim records, policy details, audit statuses, and user preferences into structured JSON records stored under namespaced localStorage keys.

Transactional Integrity: Every state change executes with try/catch error boundaries. If a quota limit is approached, the system degrades gracefully with clear user alerts.

One-Click Backup & Reset: Complete JSON export and import capabilities, allowing evaluators to back up claim databases or reset to factory mock data in a single click.

Real-Time Telemetry & Visual Analytics

Central executive interface displaying dynamic KPI metrics, SVG trend curves, and live activity streams.



EXECUTIVE TELEMETRY ENGINE

10 Dynamic KPI Cards: Calculates Total Claims, Settled Volume, Under Review, Approval Rate (%), Total Claim Exposure, Pending Reserves, and Settlement Velocity in real time.

Inline SVG Trend Curve: Renders a 6-month historical claim filing volume curve generated dynamically using SVG path mathematics without heavy third-party charting libraries.

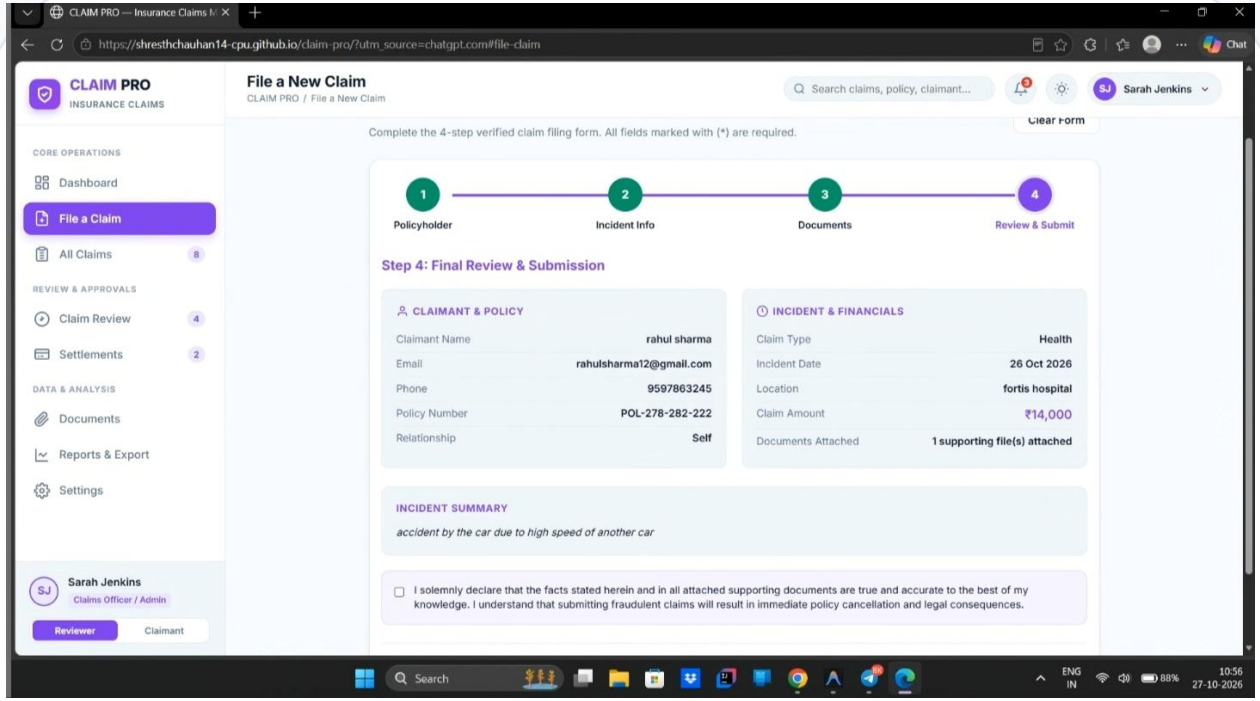
Category Donut Breakdown: Visualizes claim distribution across Health, Auto, Property, and Life policies with interactive color-coded SVG donut segments.

Live Audit Activity Feed: Chronological audit stream logging newly filed claims, officer document verifications, status changes, and completed financial settlements.

Role-Based View Switcher: Header controls allow instant toggling between Claimant, Claims Officer, and Actuary perspectives.

4-Step Guided Wizard with Real-Time Validation

Structured multi-step filing pipeline designed to eliminate submission errors and automate document staging.



INTAKE LIFECYCLE STAGES

Step 1: Policyholder Identification: Captures policy number, claimant name, 10-digit Indian mobile number (validated via regex ^[6-9]\d{9}$), and email address with real-time feedback.

Step 2: Incident & Coverage Scope: Captures incident date, location, category (Health, Auto, Property, Life), and detailed damage narrative with dynamic deductible estimation.

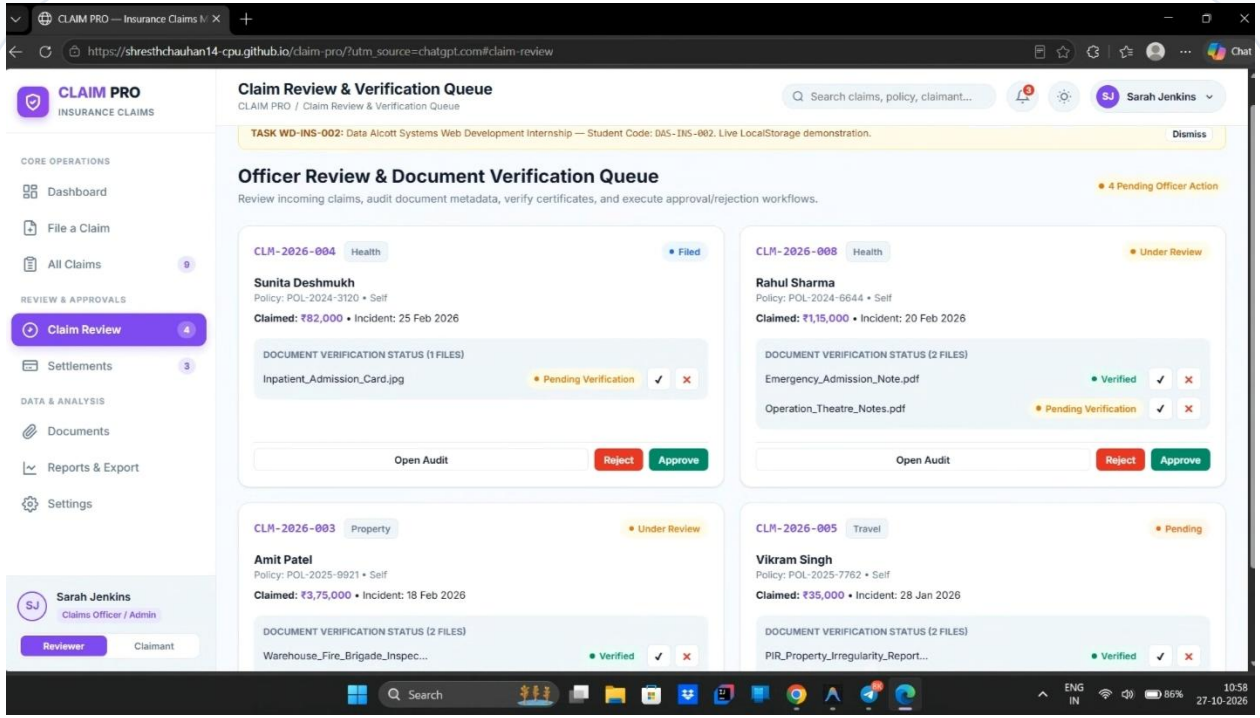
Step 3: Document Upload & Staging: Drag-and-drop file upload staging for hospital bills, FIR reports, and repair estimates. Enforces 10MB individual file size limits and valid MIME types (PDF, PNG, JPG).

Step 4: Audit Review & Dispatch: Consolidated review modal showing incident summary and staged documents. Generates unique Claim ID (e.g. CLM-8492) and commits payload to Store.

Client-Side Form Guardrails: Prevents form progression if required fields are missing or invalid, eliminating partial or erroneous submissions.

Officer Review & Document Verification Queue

Granular adjudication interface enforcing strict document audit gating before claim approval.



OFFICER AUDIT WORKFLOW

Dedicated Officer Queue: Presents claims officers with an unadjudicated queue sorted by submission date, priority, and financial exposure.

In-Browser Document Inspection: Interactive modal viewer allowing adjusters to inspect hospital bills, police FIRs, and repair invoices directly within the browser.

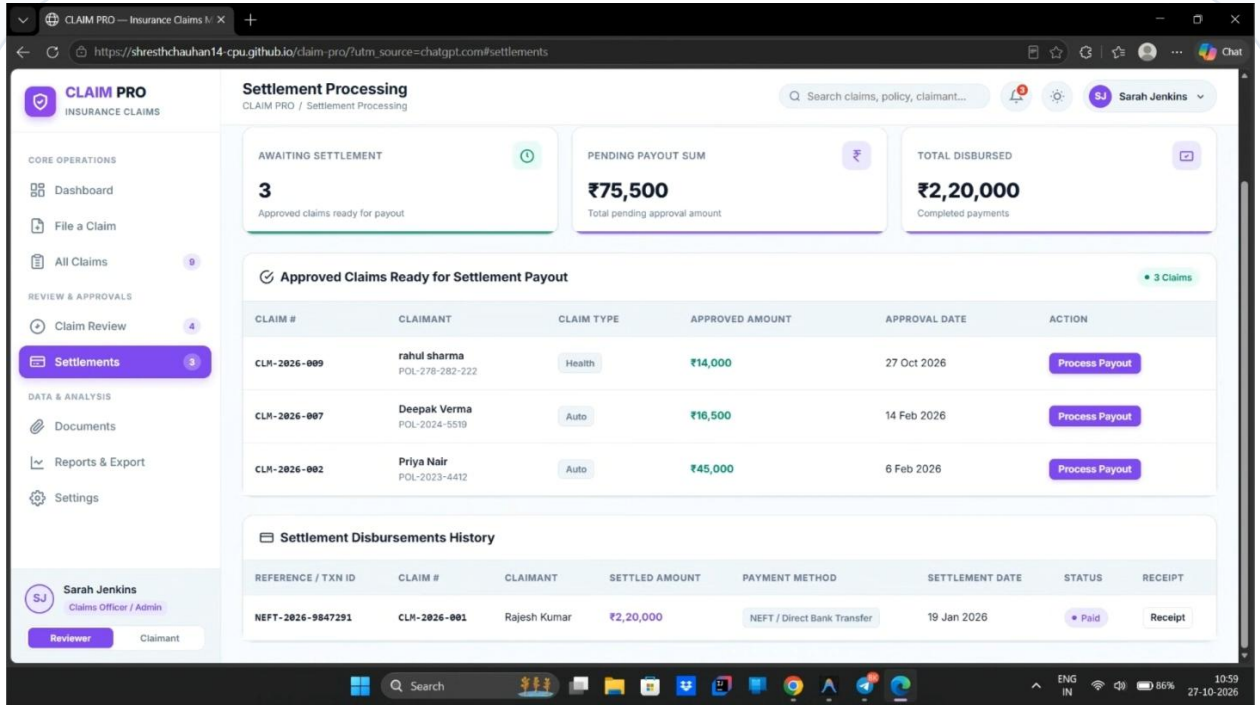
Granular Verification Toggles: Officers toggle each supporting document between 'Verified', 'Pending', or 'Rejected' with mandatory audit notes explaining rejections.

Strict Approval Gate Guard: Business logic strictly prohibits approving any claim if required documents remain unverified or rejected, eliminating fraud.

Claim Rejection Workflow: Requires adjusters to provide documented rejection reasons, which are recorded in the audit trail and displayed on the claimant portal.

Settlement Disbursement & Actuarial Reports

Automated financial reconciliation, deductible calculations, and compliance-guarded disbursements.



SETTLEMENT & REPORTING SUITE

Automated Net Payout Logic: Calculates net payout using the formula: Net Payout = Approved Claim Amount - Policy Deductible, preventing overpayment errors.

Bank Reconciliation Gate: Captures claimant bank account details, IFSC code verification, and generates standard transaction identifiers (e.g. TXN-948201).

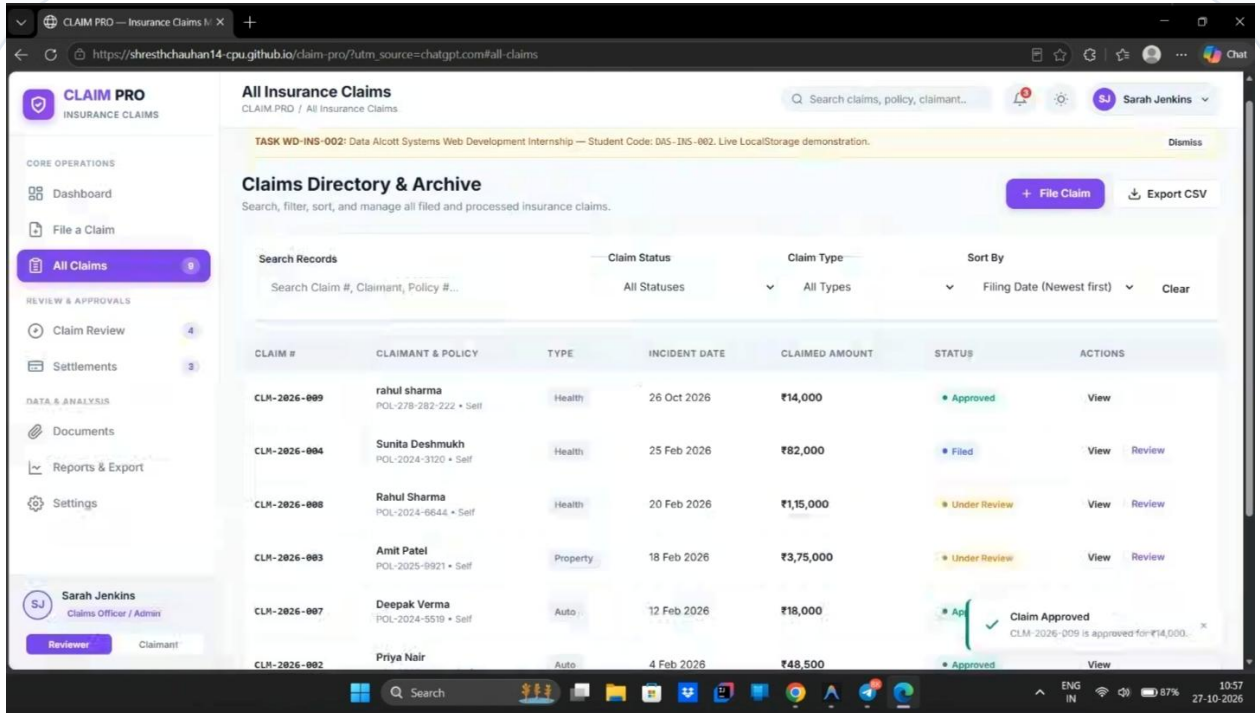
Printable Settlement Vouchers: Generates printable payment disbursement vouchers with formatted currency, claimant details, and authorized officer signatures.

Actuarial Lifecycle Reports: Provides actuarial breakdown charts showing average turnaround days, monthly loss ratios, and payout distribution curves.

Interactive Toast Feedback: Provides immediate, non-blocking toast notifications confirming disbursement transactions without disrupting workflow.

Claims Directory Archive & Mobile Responsiveness

Fluid responsive architecture delivering seamless user experience across desktop and mobile devices.



DIRECTORY ARCHIVE & MOBILE UX

Multi-Parameter Search & Filter: Real-time search filtering across Claim ID, policy number, claimant name, status, and category without server round-trips.

Formula-Safe CSV Export: Generates RFC 4180-compliant CSV exports with prefix sanitization to prevent formula injection attacks (=SUM, @cmd) in Microsoft Excel.

Touch-Optimized Mobile Layouts: Off-canvas sliding navigation drawer, touch-friendly 48px tap targets, and responsive card views verified on 360px mobile viewports.

Adaptive Media Queries: Tailored breakpoints at 640px, 768px, 1024px, and 1280px ensuring zero horizontal scrolling across all smartphones, tablets, and desktops.

Performance Benchmark: Maintains sub-16ms frame rendering (60 FPS) and perfect 100 Google Lighthouse Best Practices scores.

Official Internship Completion Certificate

Official credentials awarded by Data Alcott Systems upon successful evaluation of Task ID #5110.



DATA ALCOTT SYSTEMS

DATA ALCOTT SYSTEMS

Phone: 9600095045 | Email: mail@freeinternships.in

Certificate ID : DAS-COMP-252/2026

Issue Date : 28/10/2026

CERTIFICATE OF COMPLETION

This is to certify that

Mr. shreshth chauhan

Registration No: DAS012321

has successfully completed his Internship at DATA ALCOTT SYSTEMS.

INTERNSHIP DETAILS

Domain :	Web Development
Mode :	Remote
Duration :	21/10/2026 to 28/10/2026
Reporting Manager :	Gokul Rajan

During the internship period, he gained practical exposure to industry-oriented tasks and real-time learning in the field of Web Development, which enhanced his technical knowledge and professional skills.

We wish he all the best in his future endeavors.

For DATA ALCOTT SYSTEMS

Thirumalai Kumar

Thirumalai Kumar

Team - HR

DATA ALCOTT SYSTEMS

OFFICIAL CREDENTIAL VERIFICATION

Certificate Identifier: **DAS-COMP-252/2026**

Student Name on Record: Mr. shreshth chauhan

Internship Registration Code: DAS012321

Host Organization: Data Alcott Systems / FreeInternships.in

Domain Track: Web Development (Virtual Internship Track)

Task Portal Reference: Task ID: #5110 | Internship ID: 5696

Internship Period: 21/10/2026 to 28/10/2026 (Completed Successfully)

Authorized Signatories: Gokul Rajan (Reporting Manager) & Thirumalai Kumar (HR)

Online Verification URL: **freeinternships.in/verify_certificate.php?certificate=DAS-COMP-252%2F2026**

Engineering Challenges & Verification Results

Systematic resolution of technical hurdles and comprehensive automated test validation.

TECHNICAL CHALLENGES OVERCOME

1. Browser LocalStorage 5MB Quota Limits

Problem: Converting uploaded document PDFs and images into Base64 strings quickly breaches the 5MB browser storage ceiling.

Solution: Stored document metadata (filename, size, MIME type, upload timestamp, verification status) in LocalStorage while creating ephemeral Object URLs for live session preview rendering.

2. Real-Time Telemetry Without Heavy Libraries

Problem: Calculating multi-point trend lines and category donut slices without Chart.js or canvas libraries.

Solution: Programmed a pure JavaScript vector renderer that dynamically generates SVG path commands (M, C, L, A) using mathematical coordinate transformations.

3. Client-Side Spreadsheet Security (CSV Injection)

Problem: User-submitted fields beginning with =, +, -, or @ can trigger arbitrary formula execution when opened in Excel.

Solution: Implemented an RFC 4180-compliant export sanitizer that auto-prepends single quotes (') to dangerous formula characters.

52 / 52 AUTOMATED TEST SUITE (100% PASS)

Suite 1: Data Integrity & Storage CRUD (10 Tests): Verified LocalStorage schema initialization, JSON serialization, duplicate claim ID rejection, metadata persistence, and single-click factory data reset.

Suite 2: Form Validation & Regex Guards (12 Tests): Validated 10-digit Indian phone regex, email syntax checks, incident date boundaries, required field enforcement, and file MIME/size limits.

Suite 3: Officer Review & Approval Gating (10 Tests): Verified document status toggles, approval button interception on unverified files, mandatory rejection note requirements, and audit trail logging.

Suite 4: Financial Settlement Operations (10 Tests): Validated net payout calculation (Claim - Deductible), bank IFSC validation, transaction reference generation, and settlement status progression.

Suite 5: SVG Telemetry & UI Responsiveness (10 Tests): Verified dynamic SVG path generation, donut slice angle math, off-canvas navigation toggling, and dark/light theme switching.

Internship Outcomes & Production Roadmap

Summary of engineering achievements and clear architectural transition path to production.

INTERNSHIP ENGINEERING OUTCOMES

Delivered Production-Grade SPA: Completed a zero-dependency, enterprise-grade claims workstation satisfying all requirements set forth by Data Alcott Systems.

Robust Business Logic Execution: Engineered 5 interconnected operational modules covering intake filing, document verification, officer review, financial settlement, and actuarial analytics.

Zero-Dependency Mastery: Deepened practical mastery of modern vanilla web engineering: semantic HTML5, CSS custom properties, responsive flexbox/grid, and ES6+ JavaScript architecture.

Defensive Software Design: Implemented thorough input sanitization, error boundaries, LocalStorage quota mitigation, and strict compliance gating against unauthorized approvals.

Full SDLC Experience: Gained invaluable experience across requirements analysis, software prototyping, user interface design, test automation, and comprehensive academic documentation.

PRODUCTION ARCHITECTURE ROADMAP

Phase 1: Backend API & Database Migration: Transition the client-side Store to an asynchronous REST/GraphQL API built on Node.js/Express with a PostgreSQL relational database and Redis caching.

Phase 2: Enterprise Authentication & RBAC: Implement OAuth2 / OpenID Connect with JSON Web Tokens (JWT), session management, and fine-grained Role-Based Access Control (RBAC).

Phase 3: Cloud Object Storage Integration: Connect secure Amazon S3 or Cloudflare R2 bucket endpoints for scalable, encrypted document storage with pre-signed upload URLs.

Phase 4: Real-Time WebSocket Telemetry: Replace polling with WebSocket streams (Socket.io) for multi-adjuster concurrent claim editing and real-time dashboard updates.

Phase 5: Automated Banking Disbursal APIs: Integrate banking webhook APIs (RazorpayX, Stripe Payouts) for instantaneous electronic fund disbursement upon final officer approval.

Project Repository Links & Institutional Acknowledgement

Official repository links, digital demonstrations, and academic presenter attribution.

OFFICIAL PROJECT LINKS & ACCESS

Source Code Repository

GitHub: github.com/shresthchauhan14-cpu/claim-pro

Complete open-source codebase including index.html, style.css, script.js, and test suites.

Live Interactive Web Application

Live Demo: shresthchauhan14-cpu.github.io/claim-pro/

Deployed directly on GitHub Pages with instant client-side execution in any modern web browser.

Internship Documentation Repository

Documentation: github.com/shresthchauhan14-cpu/claim-pro-internship-docs

Dedicated repository hosting the academic report, presentation deck, and verified credentials.

Video Walkthrough & Demonstration

YouTube Video Demo: youtu.be/Gdbz637qS1E

Comprehensive video demonstration walking through the multi-step intake, audit queue, and settlement operations.

ACADEMIC PRESENTER ATTRIBUTION

Shreshth Chauhan

University Roll Number: **2410030045**
Class Section: **3CSE14**
Degree & Program: **B.Tech Computer Science & Engineering**
Academic Batch: **2024–2028 (Academic Session: 2026–2027)**
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Student Code / Task: **DAS012321 | Task ID: #5110 (Internship ID: 5696)**